



Roots of Polynomials · Worksheet 1

Core Pure 1 Chapter 4: Roots of Polynomials. Answer all questions. Show your working clearly. Total: 63 marks. Time: 80 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are three sections, one for each topic in Chapter 4. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- The roots of an equation are denoted by α , β , γ (and δ). Give exact answers, as fractions where necessary, and give equations with integer coefficients.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Sums and products of roots	19	
B Symmetric functions of the roots	20	
C Transforming equations	19	
Roar-level challenge	5	
Total	63	

Section A Sums and products of roots

19 marks

1. The roots of the equation $2x^2 - 5x + 7 = 0$ are α and β . [2]
Write down the values of $\alpha + \beta$ and $\alpha\beta$.

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2. The roots of the equation $3x^3 - 6x^2 + 4x + 9 = 0$ are α , β and γ . [3]
Write down the values of $\alpha + \beta + \gamma$, $\alpha\beta + \beta\gamma + \gamma\alpha$ and $\alpha\beta\gamma$.

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3. The roots of the equation $x^4 - 3x^3 + 2x^2 - x + 5 = 0$ are α , β , γ and δ . [3]
Write down the values of $\sum \alpha$, $\sum \alpha\beta$, $\sum \alpha\beta\gamma$ and $\alpha\beta\gamma\delta$.

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4. The roots of the equation $x^3 + px^2 + qx - 48 = 0$ are α , 2α and 3α , where p and q are real constants. [4]
Find the value of α , and hence find the values of p and q .

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5. The roots of the equation

$$x^3 - 9x^2 + kx - 15 = 0,$$

where k is a real constant, form an arithmetic sequence. The roots may be written as $a - d$, a and $a + d$.

(a) Show that one root of the equation is 3.

[2]

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(b) Find the other two roots.

[3]

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(c) Find the value of k .

[2]

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(Total for Question 5 is 7 marks)

**Section B Symmetric functions of the roots**

20 marks

6. The roots of the equation $x^2 - 4x + 7 = 0$ are α and β . Find the value of $\alpha^2 + \beta^2$. [2]

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7. For the equation in Question 6, find the values of [3]

(a) $\frac{1}{\alpha} + \frac{1}{\beta}$,

(b) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$.

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8. The roots of the equation $x^3 - 2x^2 + 3x - 5 = 0$ are α , β and γ . [3]
Find the value of $\alpha^2 + \beta^2 + \gamma^2$, and explain what your answer shows about the roots.

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9. The roots of the equation $2x^3 + x^2 - 4x + 6 = 0$ are α , β and γ . Find the values of [4]

(a) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$,

(b) $\frac{1}{\alpha\beta} + \frac{1}{\beta\gamma} + \frac{1}{\gamma\alpha}$.

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10. The roots of the equation

$$x^3 + 2x^2 - 3x + 5 = 0$$

are α , β and γ . Without solving the equation, find the value of

(a) $\alpha^2 + \beta^2 + \gamma^2$, [3]

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(b) $\alpha^3 + \beta^3 + \gamma^3$, [3]

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(c) $(\alpha + 1)(\beta + 1)(\gamma + 1)$. [2]

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(Total for Question 10 is 8 marks)



Section C Transforming equations

19 marks

11. The roots of the equation $x^2 - 3x + 5 = 0$ are α and β . [2]

Find a quadratic equation whose roots are 2α and 2β .

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12. The roots of the equation $x^3 - 4x^2 + x + 2 = 0$ are α , β and γ . [3]

Use a suitable substitution to find a cubic equation whose roots are $\alpha + 1$, $\beta + 1$ and $\gamma + 1$.

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13. The roots of the equation $2x^3 - x^2 + 3x - 1 = 0$ are α , β and γ . [3]

Find a cubic equation whose roots are $\frac{1}{\alpha}$, $\frac{1}{\beta}$ and $\frac{1}{\gamma}$.

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14. The roots of the equation $x^3 + 3x^2 - 2x + 1 = 0$ are α , β and γ . [4]

Use the substitution $y = x^2$ to find a cubic equation whose roots are α^2 , β^2 and γ^2 .

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15. The roots of the equation

$$2x^3 + 5x^2 - 3x + 4 = 0$$

are α , β and γ .

- (a) Use a substitution to find a cubic equation, with integer coefficients, whose roots are $(2\alpha - 1)$, $(2\beta - 1)$ and $(2\gamma - 1)$. [4]

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- (b) Hence find the value of [3]

- (i) $(2\alpha - 1)(2\beta - 1)(2\gamma - 1)$,
(ii) $(2\alpha - 1)^2 + (2\beta - 1)^2 + (2\gamma - 1)^2$.

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(Total for Question 15 is 7 marks)



ROAR-LEVEL CHALLENGE

5 marks

16. The roots of the equation $x^3 + 2x^2 - x + 3 = 0$ are α , β and γ . [5]
Find a cubic equation, with integer coefficients, whose roots are $\alpha\beta$, $\beta\gamma$ and $\gamma\alpha$.

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END OF WORKSHEET

Total: 63 marks